Lecture 13

Oriented geometry and work

Uses.

Summary 2: Standing fact E supplies the trigonometric identities and radians.
Summary 6: The proved limits are $\sin h/h \to 1$ and $(\cos h - 1)/h \to 0$ as $h \to 0$. Finite sums preserve limits.
Summary 12: Vector operations are componentwise. The dot product is bilinear, defines the norm and the angle, and gives orthogonal projection. Cauchy–Schwarz has equality exactly for dependent pairs. Independence means that a zero linear combination has only zero coefficients.

Objectives. *The dot product measures a component along a direction. The cross product records an area together with a choice of normal direction. We use these products in stated models for work and torque, then calculate signed volumes. Their signs depend on a fixed orientation of the coordinate axes.*

By the end of this lecture you will be able to:

- Prove the cross product's orthogonality and area formula in right-handed coordinates.
- Calculate constant-force work and torque under the stated physical models.
- Prove the scalar triple-product criterion for coplanarity and compute signed volumes.
- Convert between parametric and normal plane equations and keep cross-product order and parentheses.

13.1 The cross product and orientation

An ordered set of three mutually perpendicular unit axes has a choice of *orientation*. The axes are *right-handed* when curling the right-hand fingers from the first positive axis towards the second makes the thumb point along the third positive axis.

Convention C. Spatial Cartesian axes are right-handed throughout this course. The cross product uses this orientation. Coordinate-angle and boundary conventions will be specified when those objects are introduced.

Definition 13.1 (Cross product). For $\boldsymbol{a} = (a_1, a_2, a_3)$ and $\boldsymbol{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$, their *cross product* is the vector

$$\boldsymbol{a} \times \boldsymbol{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1).$$

This definition concerns three-dimensional vectors. It is bilinear because every component is linear in each argument. Interchanging the two arguments negates each component, so $\boldsymbol{b} \times \boldsymbol{a} = -\boldsymbol{a} \times \boldsymbol{b}$. This property is called *antisymmetry*. In particular, $\boldsymbol{a} \times \boldsymbol{a} = \boldsymbol{0}$. The standard coordinate vectors satisfy

$$\boldsymbol{e}_1 \times \boldsymbol{e}_2 = \boldsymbol{e}_3, \qquad \boldsymbol{e}_2 \times \boldsymbol{e}_3 = \boldsymbol{e}_1, \qquad \boldsymbol{e}_3 \times \boldsymbol{e}_1 = \boldsymbol{e}_2.$$

Reversing any displayed pair gives the negative vector. These identities specify the sign of the normal direction in the chosen axes.

Theorem 13.2 (Orthogonality and magnitude). *For $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$, the vector $\boldsymbol{a} \times \boldsymbol{b}$ is orthogonal to both $\boldsymbol{a}$ and $\boldsymbol{b}$, and*

$$\|\boldsymbol{a} \times \boldsymbol{b}\|^2 = \|\boldsymbol{a}\|^2 \, \|\boldsymbol{b}\|^2 - (\boldsymbol{a} \cdot \boldsymbol{b})^2. \tag{13.1}$$

The cross product vanishes exactly when the pair is linearly dependent. For two nonzero vectors with angle $\theta \in [0, \pi]$,

$$\|\boldsymbol{a} \times \boldsymbol{b}\| = \|\boldsymbol{a}\| \, \|\boldsymbol{b}\| \sin\theta.$$

Proof. The first orthogonality calculation is

$$\boldsymbol{a} \cdot (\boldsymbol{a} \times \boldsymbol{b}) = a_1 a_2 b_3 - a_1 a_3 b_2 + a_2 a_3 b_1 - a_2 a_1 b_3 + a_3 a_1 b_2 - a_3 a_2 b_1 = 0.$$

The terms cancel in pairs. Antisymmetry gives $\boldsymbol{b} \cdot (\boldsymbol{a} \times \boldsymbol{b}) = -\boldsymbol{b} \cdot (\boldsymbol{b} \times \boldsymbol{a}) = 0$ by the same established calculation with the two vectors interchanged. For the squared magnitude, the three squared components give

$$\sum_{1 \le i < j \le 3} (a_i b_j - a_j b_i)^2 = \sum_{i \ne j} a_i^2 b_j^2 - 2 \sum_{i < j} a_i a_j b_i b_j.$$

Expanding $(\sum_i a_i^2)(\sum_j b_j^2) - (\sum_i a_i b_i)^2$ cancels the three terms $a_i^2 b_i^2$ and gives precisely the same expression. This proves (13.1).

The squared magnitude is zero exactly in the equality case of Cauchy–Schwarz, which is dependence of the pair. For nonzero vectors, the angle formula makes the right side $\|\boldsymbol{a}\|^2 \, \|\boldsymbol{b}\|^2 \, (1 - \cos^2\theta)$. Since $\sin\theta \ge 0$ on $[0, \pi]$, its nonnegative square root is the stated magnitude. ∎

The *parallelogram* with edges $\boldsymbol{a}$ and $\boldsymbol{b}$ from the origin consists of $s\boldsymbol{a} + t\boldsymbol{b}$ with $0 \le s, t \le 1$. For nonzero edges, its elementary base-times-height area is $\|\boldsymbol{a}\| \, \|\boldsymbol{b}\| \sin\theta$. Indeed, projecting $\boldsymbol{b}$ onto $\boldsymbol{a}$ leaves an orthogonal height of length $\|\boldsymbol{b}\| \sin\theta$ by Lecture 12's Pythagorean identity. Thus the parallelogram area is $\|\boldsymbol{a} \times \boldsymbol{b}\|$. If the edges are dependent, the height and area are zero. The triangle with the same two edges has half this area. The cross product's magnitude gives area. For an independent ordered pair, its *oriented unit normal* is $(\boldsymbol{a} \times \boldsymbol{b})/\|\boldsymbol{a} \times \boldsymbol{b}\|$ under convention C. In the $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ coordinate plane, set $r = \|\boldsymbol{a}\| > 0$ and $s = \|\boldsymbol{b}\| > 0$ and write $\boldsymbol{a} = (r\cos\varphi, r\sin\varphi, 0)$ and $\boldsymbol{b} = (s\cos\psi, s\sin\psi, 0)$. Such angles exist: choose $\arccos(a_1/r)$ or its reflection $2\pi - \arccos(a_1/r)$ according to the sign of a_2. The sine-square identity gives the required second component. The same construction applies to $\boldsymbol{b}$. The addition formula gives

$$\boldsymbol{a} \times \boldsymbol{b} = (0, 0, rs\sin(\psi - \varphi)).$$

For an independent pair, choose the shorter signed turn from $\boldsymbol{a}$ to $\boldsymbol{b}$ in $(-\pi, \pi)$. Its sine is positive for a counterclockwise turn and negative for a clockwise turn. The normal therefore points along $\boldsymbol{e}_3$ in the first case and $-\boldsymbol{e}_3$ in the second.

Figure 13.1 shows the perpendicular height and normal sign.

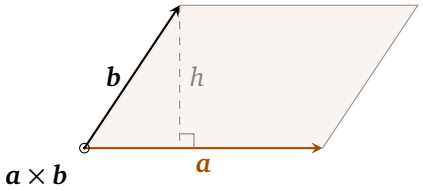


Figure 13.1: A nondegenerate parallelogram in the (e_1, e_2) coordinate plane, with height h. The dot in a circle denotes the normal pointing out of the page when the turn from a to b is counterclockwise.

Example 13.3 (Area from components). Take $a = (2, 1, 0)$ and $b = (0, 1, 2)$. Their cross product is

$$a \times b = (1 \cdot 2 - 0 \cdot 1,\ 0 \cdot 0 - 2 \cdot 2,\ 2 \cdot 1 - 1 \cdot 0) = (2, -4, 2).$$

Its dot products with the edges are $4 - 4 = 0$ and $-4 + 4 = 0$. Its squared norm is $4 + 16 + 4 = 24$. The parallelogram therefore has area $2\sqrt{6}$, and the triangle has area $\sqrt{6}$, in squared coordinate-length units. Reversing the edges changes the normal to $(-2, 4, -2)$ and preserves the area.

13.2 Work and torque in a stated force model

A *force vector* specifies a force's components, measured in newtons. We use the following mechanical models as physical premises. For a constant force F and a net displacement d, the work is

$$W = F \cdot d. \tag{13.2}$$

Work is a scalar, measured in joules when displacement is in metres. One joule is one newton metre. For nonzero displacement, this is the signed force component in its direction, $F \cdot (d/\|d\|)$, multiplied by $\|d\|$. Equivalently, for two nonzero vectors the work is $\|F\|\,\|d\| \cos\theta$. Work is negative when the force component opposes the displacement. A zero force or zero displacement gives zero work without defining an angle. The constant-force endpoint formula is the model here. Lecture 40 treats forces that vary along a path.

Fix a reference point O. Let r be the displacement from O to the point where a force F acts. The *torque about* O in the force model is

$$\tau = r \times F. \tag{13.3}$$

Torque is a vector measured in newton metres. It depends on the chosen reference point. For nonzero r and F, its magnitude is $\|r\|\,\|F\| \sin\theta$. The *line of action* of a nonzero force is the line through its application point in direction F. The distance from O to this line is the *lever arm*. Lecture 12's projection decomposition gives its length as $\|r\| \sin\theta$. Thus torque magnitude is force magnitude times lever arm. If $r = 0$, the lever arm and torque are zero. A force whose line of action goes through O also has zero torque.

Figure 13.2 shows the lever arm in the plane of $\boldsymbol{r}$ and $\boldsymbol{F}$. The perpendicular segment ends on the line of action, which may extend behind the application point.

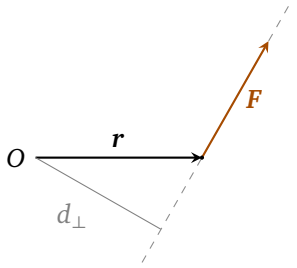


Figure 13.2: The lever arm $d_\perp$ is the perpendicular distance from the reference point to the force's line of action. Its length, multiplied by force magnitude, gives torque magnitude.

Example 13.4 (The signs of work and torque)**.** In a chosen set of right-handed axes, let a constant force be $\boldsymbol{F} = (2, -1, 3)$ newtons. For the displacement $\boldsymbol{d} = (3, 1, -2)$ metres, its work is

$$W = 2 \cdot 3 + (-1) \cdot 1 + 3 \cdot (-2) = -1 \text{ J}.$$

The force therefore does negative work over this displacement. At an instant when the force acts at $\boldsymbol{r} = (1, 2, -1)$ metres relative to O, its torque is

$$\boldsymbol{\tau} = (2 \cdot 3 - (-1)(-1),\ (-1) \cdot 2 - 1 \cdot 3,\ 1 \cdot (-1) - 2 \cdot 2) = (5, -5, -5) \text{ N m}.$$

The torque magnitude is $5\sqrt{3}$ newton metres. The signed torque component along the positive third axis is -5 newton metres. That component records the sense of the tendency to turn about this axis. The work and torque refer to different data: net displacement for work, and the position of application relative to O for torque.

Torque and a small rotation

The sign of an axial torque agrees with the work over a small rotation. Keep a constant force $\boldsymbol{F} = (F_1, F_2, F_3)$ and an initial application position $\boldsymbol{r} = (x, y, z)$, with $x, y, z \in \mathbb{R}$. A positive rotation by φ radians about the third axis has position

$$(x \cos\varphi - y \sin\varphi,\ x \sin\varphi + y \cos\varphi,\ z).$$

This is the usual planar rotation in the first two coordinates. Subtracting the initial position gives the displacement $\Delta\boldsymbol{r}$. The standard limits from Lecture 6 yield

$$\frac{\Delta\boldsymbol{r}}{\varphi} \longrightarrow (-y, x, 0) \qquad (\varphi \to 0),$$

where a vector limit here means the limit of each of its components. Since the force is constant, the work model gives

$$\frac{W}{\varphi} = \boldsymbol{F} \cdot \frac{\Delta\boldsymbol{r}}{\varphi} \longrightarrow -yF_1 + xF_2 = (\boldsymbol{r} \times \boldsymbol{F}) \cdot \boldsymbol{e}_3.$$

This derives the axial torque as the limiting work per unit rotation in this model. For the worked force and application point, this limit is $-2 \cdot 2 + 1 \cdot (-1) = -5$ newton metres, agreeing with the third component above. No assumption about a body's mass or resulting motion was used.

13.3 Signed volume and coplanarity

The *scalar triple product* of vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ in $\mathbb{R}^3$ is

$$[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = (\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}. \tag{13.4}$$

It is a scalar. Expanding the component definition gives

$$[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_2 b_1 c_3 - a_3 b_2 c_1.$$

This expression is linear in each argument, hence *trilinear*. A cyclic change of the arguments preserves it:

$$[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = [\boldsymbol{b}, \boldsymbol{c}, \boldsymbol{a}] = [\boldsymbol{c}, \boldsymbol{a}, \boldsymbol{b}].$$

Indeed, a cyclic change only reorders each group of three terms. Interchanging two arguments exchanges the positive and negative terms, so it negates the product. A repeated argument therefore gives zero. In particular, the product also equals $\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})$.

The *parallelepiped* with these edges consists of $s\boldsymbol{a} + t\boldsymbol{b} + u\boldsymbol{c}$ with $0 \le s, t, u \le 1$. If $\boldsymbol{a} \times \boldsymbol{b} \neq \boldsymbol{0}$, its base has area $\|\boldsymbol{a} \times \boldsymbol{b}\|$. The unit normal to the base is $\boldsymbol{n} = (\boldsymbol{a} \times \boldsymbol{b})/\|\boldsymbol{a} \times \boldsymbol{b}\|$. Its height is the absolute scalar component $|\boldsymbol{c} \cdot \boldsymbol{n}|$. The base-times-height formula therefore gives the volume

$$V = \|\boldsymbol{a} \times \boldsymbol{b}\| \; |\boldsymbol{c} \cdot \boldsymbol{n}| = |[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}]|.$$

If the base edges are dependent, both the volume and the triple product are zero. The *signed volume* is the triple product itself. Its sign says which side of the oriented base contains the third edge's tip.

Figure 13.3 shows a base and its height in an oblique view. The third edge need not be normal to the base.

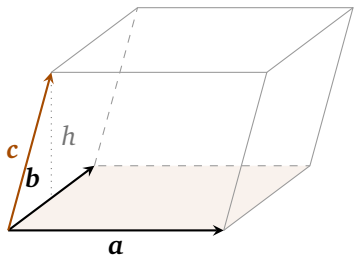


Figure 13.3: An oblique view of the parallelepiped. The height h is the absolute normal component of $\boldsymbol{c}$ above the shaded base. This projection does not preserve right angles.

Example 13.5 (Volume over the previous base). Keep $\boldsymbol{a} = (2, 1, 0)$ and $\boldsymbol{b} = (0, 1, 2)$, and take $\boldsymbol{c} = (1, -1, 3)$. The signed volume is

$$[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = (2, -4, 2) \cdot (1, -1, 3) = 2 + 4 + 6 = 12.$$

Its volume is 12 cubic coordinate-length units. The height above the base is $12/(2\sqrt{6}) = \sqrt{6}$. Interchanging the first two edges gives signed volume -12 and the same ordinary volume.

Vectors are *coplanar* if they can all be expressed as linear combinations of at most two fixed vectors. This includes a line and the zero vector. The plane interpretation is established in § 13.4.

Theorem 13.6 (Coplanarity criterion). *Three vectors in $\mathbb{R}^3$ are coplanar exactly when their scalar triple product is zero.*

Proof. Suppose the three vectors are linear combinations of two vectors $\boldsymbol{p}$ and $\boldsymbol{q}$. Trilinearity expands their triple product into terms with three arguments chosen from these two vectors. Every term has a repeated argument and is zero. Thus coplanarity implies a zero triple product.

Conversely, suppose $[\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = 0$. If $\boldsymbol{a}$ and $\boldsymbol{b}$ are dependent, all three vectors are combinations of $\boldsymbol{c}$ and a nonzero member of the first pair. If both members of the pair are zero, $\boldsymbol{c}$ alone suffices. Thus the vectors are coplanar in this case.

Suppose $\boldsymbol{a}$ and $\boldsymbol{b}$ are independent. Their cross product is nonzero, so at least one of its components is nonzero. Under the cyclic coordinate change $1 \mapsto 2 \mapsto 3 \mapsto 1$, the cross-product components cycle in the same way and the triple-product expansion is unchanged. Thus a cyclic relabelling can place a nonzero component third. In those coordinates, take $D = a_1 b_2 - a_2 b_1 \neq 0$. Define

$$\alpha = \frac{c_1 b_2 - c_2 b_1}{D}, \qquad \beta = \frac{a_1 c_2 - a_2 c_1}{D}.$$

Substitution gives $\alpha a_1 + \beta b_1 = c_1$ and $\alpha a_2 + \beta b_2 = c_2$. For the third component, expanding the numerator gives

$$D(c_3 - \alpha a_3 - \beta b_3) = c_1(a_2 b_3 - a_3 b_2) + c_2(a_3 b_1 - a_1 b_3) + c_3(a_1 b_2 - a_2 b_1) = [\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}] = 0.$$

Since $D \neq 0$, the third component also agrees. Hence $\boldsymbol{c} = \alpha \boldsymbol{a} + \beta \boldsymbol{b}$ and all three are coplanar. ∎

13.4 Planes and their normal vectors

A *plane through a point* with position $\boldsymbol{p}$ and independent directions $\boldsymbol{u}$ and $\boldsymbol{v}$ consists of the positions

$$\boldsymbol{x} = \boldsymbol{p} + s\boldsymbol{u} + t\boldsymbol{v}, \qquad s, t \in \mathbb{R}. \tag{13.5}$$

These are its *parametric equations*. The two real numbers s and t are independent parameters. Independence of the directions is necessary: a dependent pair would describe at most a line. This also explains the geometric meaning of coplanarity. Independent generators give a plane through the origin. A family with just one nonzero generating direction lies in a plane obtained by adding a standard coordinate vector not parallel to it. Such a vector exists because $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ are not parallel. The all-zero family lies in the plane generated by $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$. Conversely, if $\boldsymbol{0} = \boldsymbol{p} + s_0\boldsymbol{u} + t_0\boldsymbol{v}$ and $\boldsymbol{x} = \boldsymbol{p} + s\boldsymbol{u} + t\boldsymbol{v}$ lie in the same plane, subtraction gives $\boldsymbol{x} = (s - s_0)\boldsymbol{u} + (t - t_0)\boldsymbol{v}$. The parameters of each point are unique. Indeed, two representations would give $(s_1 - s_2)\boldsymbol{u} + (t_1 - t_2)\boldsymbol{v} = \boldsymbol{0}$, forcing both differences to vanish by independence.

A *normal vector* to this plane is a nonzero vector orthogonal to both directions. The cross product $\boldsymbol{n} = \boldsymbol{u} \times \boldsymbol{v}$ is such a normal. Every point in the plane satisfies the *normal equation*

$$\boldsymbol{n} \cdot (\boldsymbol{x} - \boldsymbol{p}) = 0. \tag{13.6}$$

Conversely, this equation says $[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{x} - \boldsymbol{p}] = 0$. The proof of Theorem 13.6, with the first pair independent, expresses $\boldsymbol{x} - \boldsymbol{p} = s\boldsymbol{u} + t\boldsymbol{v}$. Thus the normal equation and the parametric

equations describe exactly the same set of points. A nonzero scalar multiple of the normal gives the same plane, while reversing the normal reverses its chosen orientation.

Every equation $\boldsymbol{n} \cdot \boldsymbol{x} = c$ with $\boldsymbol{n} \in \mathbb{R}^3 \setminus \{\mathbf{0}\}$ and $c \in \mathbb{R}$ describes a plane. To see this directly, first choose a nonzero component of $\boldsymbol{n}$. Use the same cyclic coordinate change as above to take $n_3 \neq 0$. A point and two independent directions satisfying the equation are

$$\boldsymbol{p} = (0, 0, c/n_3), \qquad \boldsymbol{u} = (1, 0, -n_1/n_3), \qquad \boldsymbol{v} = (0, 1, -n_2/n_3).$$

The directions are independent because their first two coordinate pairs are $(1, 0)$ and $(0, 1)$. A point on the normal equation obeys $x_3 = c/n_3 - (n_1/n_3)x_1 - (n_2/n_3)x_2$. Hence it equals $\boldsymbol{p} + x_1\boldsymbol{u} + x_2\boldsymbol{v}$. Conversely, substitution shows every such parametric point satisfies the normal equation.

Example 13.7 (Two equations for one plane). Take the point $\boldsymbol{p} = (1, -1, 2)$ and the previous directions $\boldsymbol{u} = (2, 1, 0)$ and $\boldsymbol{v} = (0, 1, 2)$. The cross product is $(2, -4, 2)$, so choose the simpler normal $\boldsymbol{n} = (1, -2, 1)$. The plane has parametric and normal descriptions

$$(x, y, z) = (1 + 2s, -1 + s + t, 2 + 2t), \qquad x - 2y + z = 5.$$

For the normal equation, the constant is $\boldsymbol{n} \cdot \boldsymbol{p} = 1 + 2 + 2 = 5$. The parameters $s = t = 1$ give the point $(3, 1, 4)$. It also satisfies the normal equation because $3 - 2 + 4 = 5$. The point $(1, 0, 2)$ does not lie in the plane, since $1 - 0 + 2 = 3 \neq 5$. Lecture 14 uses this plane in an intersection and distance problem.

13.5 Order and parentheses

A cross product produces a vector, so another cross product can be taken. The placement of parentheses matters. For instance,

$$\boldsymbol{e}_1 \times (\boldsymbol{e}_1 \times \boldsymbol{e}_2) = -\boldsymbol{e}_2, \qquad (\boldsymbol{e}_1 \times \boldsymbol{e}_1) \times \boldsymbol{e}_2 = \mathbf{0}.$$

The first equality uses $\boldsymbol{e}_1 \times \boldsymbol{e}_2 = \boldsymbol{e}_3$ and $\boldsymbol{e}_1 \times \boldsymbol{e}_3 = -\boldsymbol{e}_2$. The second uses $\boldsymbol{e}_1 \times \boldsymbol{e}_1 = \mathbf{0}$. Thus the cross product is not associative. A scalar triple product and an iterated cross product are also different types of object: the former is a scalar, while the latter is a vector. Sheet 13 supplies the full component proof of

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b}).$$

No preceding proof in this lecture uses this additional identity.

Summary 13.

- Definition 13.1 and Theorem 13.2: in right-handed axes, the cross product is bilinear and antisymmetric, is orthogonal to its two arguments, and has magnitude equal to their parallelogram area. It vanishes exactly for a dependent pair.
- Equations (13.2) and (13.3): the stated models give constant-force work $\boldsymbol{F} \cdot \boldsymbol{d}$ and torque about a reference point $\boldsymbol{r} \times \boldsymbol{F}$. The axial torque is the limiting work per unit small rotation about that axis, as proved for the third coordinate axis.
- Equation (13.4) and Theorem 13.6: the scalar triple product is a signed volume. It is cyclically invariant, changes sign under an exchange, and vanishes exactly for coplanar vectors.
- Equations (13.5) and (13.6): independent plane directions give a nonzero cross-product normal. Parametric and normal equations describe the same plane.
- Cross products require their argument order and parentheses. The cross product is not associative.

Sheet 13 practices areas, mechanical signs and degenerate volumes, then proves the additional vector identity. Lecture 14 uses normal directions to describe lines, planes and their intersections.

References

J. Stewart, *Calculus: Early Transcendentals*, 8th ed., Cengage, 2016, §§12.3–12.4. K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, §§7.6–7.7.